

Patrick Wagstrom

Data Engineering and Artificial Intelligence Leader

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Strategic and hands-on executive leader in artificial intelligence, machine learning, data engineering, and software development with extensive experience in organizations across finance, telecom, and video streaming. Proven track record in driving impactful AI/ML initiatives, building high-performing teams, modernizing data pipelines, and driving significant business value.

Professional Experience

Grainger, Chicago, IL and Hartford, CT

Senior Director Applied ML, July 2023 - August 2024

- Led team of 43 FTE ML scientists, software engineers, and product managers on product search and discovery, LLM agents, ML in mobile applications, data pipelines, and governance.
- Developed an LLM-based customer service agent using search and unstructured data with OpenAI on Azure to provide human-in-the-middle assistance to Grainger employees.
- Enhanced the search capabilities of Grainger.com, the 11th largest ecommerce site in the US, through introduction of vector and hybrid search.
- Designed initial version overall data architecture - integrating an S3 data lake with data products, Kafka event streaming, Databricks for ML and data processing workflows, Snowflake for analytical workflows, and Atlan for data governance.
- Led the effort to define the ML Scientist job family across the company and integrate it into overall talent strategy including recruiting and talent development.
- Worked with legal partners to define and implement generative AI governance process.

Brightcove, Boston, MA and Hartford, CT

Chief Data Officer, May 2021 - December 2022

- Responsible for all analytics, data, and machine learning at Brightcove - including engineering, governance, policy, and vendor relations.
- Owned end-to-end analytics collection and processing systems handling trillions of rows of data and billions of monthly video views on both AWS and GCP.
- Managed an organization of 34 FTEs and 12 contractors, overseeing data platforms, analytics and insights, data and model governance, and customer-facing data products.
- Executed the \$12.3 million acquisition of Wicket Labs, expanding Brightcove's viewer analytics capabilities and growing the product customer base by over 50x in under a year.
- Collaborated closely with C-level executives at Brightcove and customers, driving corporate strategies and fostering strong business relationships.
- Negotiated and owned a multi-million dollar partnership with Google Cloud Platform.
- Maintained and expanded multi-cloud data infrastructure using tools such as Trino, Glue, Lambda, SageMaker, and Redshift on AWS and BigQuery, Cloud Composer, Dataflow, and Pub/Sub on GCP.

Verizon, Basking Ridge, NJ and Hartford, CT

Director of Emerging Technology, September 2019 - April 2021

- Built and managed a team of 9 FTEs and 51 vendor contractors with a \$15 million annual budget, collaborating closely with CIO, CTO, and CDAO to influence corporate strategies.
- Developed cell site energy efficiency machine learning models, resulting in \$5 million annual savings and the creation of a digital twin for future simulation.
- Led the development of a holistic model connecting online brand discussions to customer profiles, supporting customer service operations and building patent-pending bot detection models.
- Designed a new data architecture to migrate from ad-hoc on-prem Teradata and Hadoop solutions to a managed streaming data architecture with Google BigQuery.
- Architected an enterprise strategy for reproducible machine learning and MLOps.
- Led the software engineering dojos - intensive collaboration environments to improve engineering efficiency - until the onset of COVID-19 halted face-to-face interactions.

- Worked alongside legal and policy experts to create several Verizon policies, including facial recognition, energy efficiency, model risk management, and data privacy.

Capital One, New York, NY and McLean, VA

Senior Director of Data Science for Machine Learning Platforms, January 2019 - September 2019

Director of Data Science for Machine Intelligence, November 2016 - January 2019

- Architect and overall lead of the Capital One Card Machine Learning Platform - a platform for scalable and personalized real-time reinforcement learning-based machine learning models using tools such as Kubernetes, gRPC, and AWS in a heavily regulated environment.
- Led a team that deployed and managed multi-modal customer intelligence models working on both structured and unstructured text and voice data, resulting in \$27 million annual savings.
- Hired, managed, and developed a distributed team of 18 data scientists and data analysts.
- Built out a \$1.2mm academic research partnership with universities to explore responsibility and fairness in artificial intelligence.
- Conducted more than 300 hiring interviews to help grow Card Machine Learning from 33 to 168 people and Capital One's New York Card team from 8 to nearly 200.
- Championed reproducible machine learning and MLOps across the organization, ensuring best practices in data management, model training and refit, model serving, and risk management.

IBM Watson, Littleton, MA

Research Staff Member/Technical Lead, January 2014 - November 2016

- Led a globally distributed team that built the IBM Watson Conversation service - an innovative service to create rich conversational interfaces with natural language processing.
- Global team lead for Watson Developer Cloud Tooling. Created cutting edge applications for creating, training, and maintaining cognitive and machine learning solutions including Watson Engagement Advisor and IBM Watson Natural Language Classifier.
- Technical lead for Chef Watson and the IBM Food Truck - an innovative generative AI solution demonstrated in 2014 garnering more than 1 billion media impressions.
- Researched, designed, and built tools to customize AI/ML models and systems at scale - supporting hundreds of thousands of users and use cases.
- Traveled around the world to conduct research with and work side-by-side with our customers to better understand the potential for AI/ML.
- Led evolution of internal development standards and tools from a legacy Java stack to a modern stack based on Node.js and embracing tools such as GitHub.

IBM TJ Watson Research Center, Yorktown Heights, NY

Research Staff Member, August 2009 - January 2014

- Team lead for the "Millennial Enterprise" aspect of IBM's 2013 Global Technology Outlook, which described the need for user data in building out AI/ML to IBM and customer C-Level executives.
- Analytics lead for JazzHub, IBM's cloud software development strategy. Designed analytics strategy, introduced A/B testing, and developed analytics dashboards.
- Published papers on topics around distributed collaboration, technical debt in software, and flow of ideas in software engineering communities.

Education

Ph.D. in Engineering and Public Policy and Computation, Organizations, and Society, May 2009

Carnegie Mellon University, Pittsburgh, PA

MS in Computation, Organizations, and Society, May 2007

Carnegie Mellon University, Pittsburgh, PA

MS in Computer Science, August 2003

Illinois Institute of Technology, Chicago, IL

BS in Computer Science / BS in Computer Engineering / BS in Electrical Engineering, May 2002

Illinois Institute of Technology, Chicago, IL